



User Guide (v1.0)

Gwanho Yoon, Ph.D.

May 6, 2020

Contents

1	Introduction	1
1.1	Layout	2
1.2	Basic Use	6
1.3	Coordinate Information	7
1.4	Exit	8
2	Materials	9
2.1	Material Import	9
2.2	Material Browser	10
3	Structures	11
3.1	Film	11
3.2	Rectangle	12
3.3	Circle	13
3.4	Mixed	14
3.5	Polygon	15
3.6	Cone	16
3.7	Horn	17
3.8	Delete	18
3.9	Index Viewer	19
4	Schedule	20
4.1	Basic Management	20
4.2	Auto Complete	21
4.3	Table Wizard	22
4.4	Reset	23
5	Execution	24
5.1	Run	24
5.2	Cancellation	25
6	Result	26
6.1	Transmittance & Reflectance	26
6.2	Diffraction Analysis	27
6.3	Data Export	28
7	Information	29
7.1	Packages	29
7.2	License	30

1 Introduction

MAXIM is free electromagnetic wave simulation software based on rigorous coupled-wave analysis (RCWA) calculating diffraction characteristics of a periodically structured multilayer system. MAXIM has several distinct advantages compared with other open-source RCWA programs. They have shown excellent computation performance in terms of accuracy and speed, but suffer from low accessibility to beginners who are not familiar with programming languages. In contrast, MAXIM benefits from extremely easy accessibility due to straightforward graphical user interfaces (GUIs) based on a spread sheet which is perhaps the most accessible format of contemporary computation programs. The main application of MAXIM is to simulate subwavelength nanostructures for dielectric metasurfaces where the physical shape of antennas has a critical function. Therefore, two dimensional gratings of almost arbitrary shapes can be considered in MAXIM drastically increasing the availability. MAXIM also provides powerful functions to build and manage an enormous computation schedule without difficulty. Such notable advantage of MAXIM is very favorable to optimize the design of metasurfaces. Therefore, considering those benefits, we believe that MAXIM will provide a real opportunity of using RCWA to anyone including beginners.

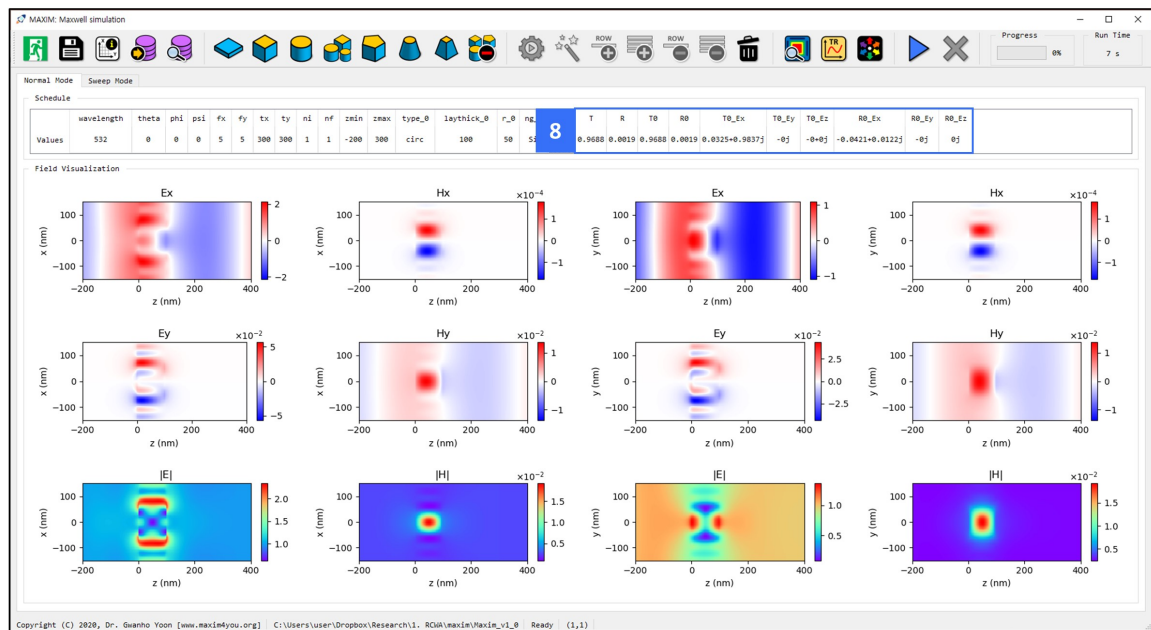
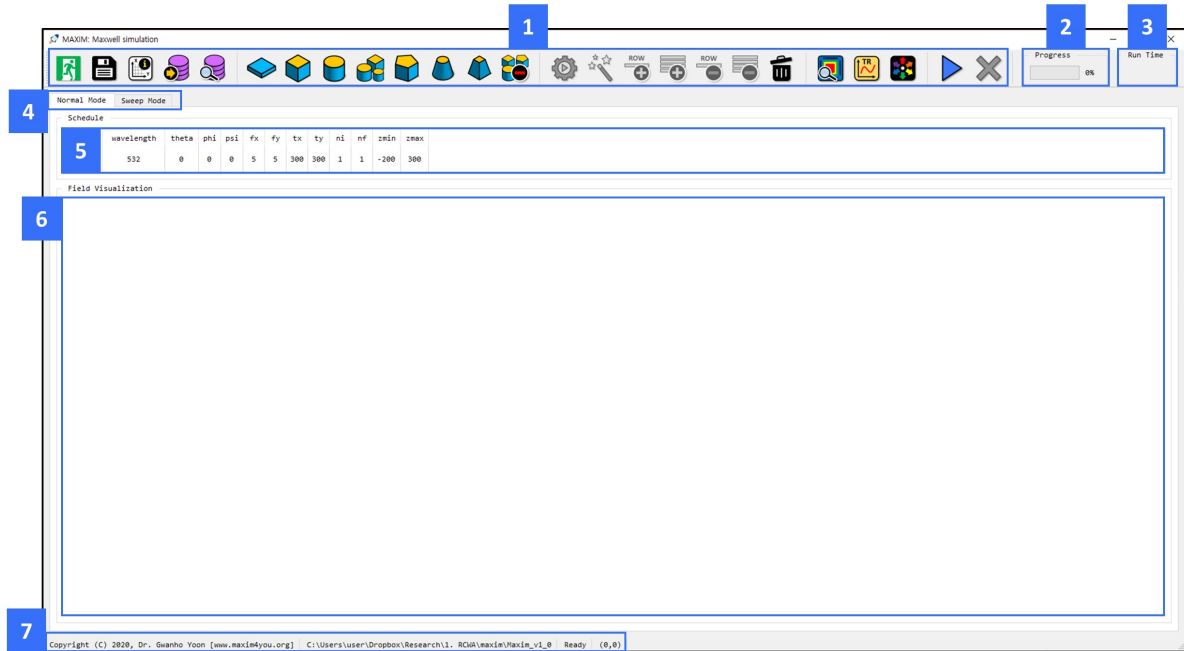
This guide describes the usage of MAXIM and its functions in detail. Section 1 gives the basic introduction of the program and its coordinate information. Section 2 explains the functions related to material properties. Section 3 elucidates the functions to create structures. Section 4 presents how to manage the computation schedule using special functions. Section 5 describes the execution and cancellation of computation. Section 6 introduces simple visualization and export of the results. Section 7 contains general information including the license of MAXIM.

To better understand the use of MAXIM, it is highly recommended to see a tutorial movie on the website (<https://ylab.seoultech.ac.kr>).

1.1 Layout

<Normal Mode>

MAXIM provides two operation modes of a Normal mode and Sweep mode. In the Normal mode, one simulation condition can be examined by one computation. The calculation results include not only transmission and reflection characteristics, but also electromagnetic field profiles throughout existing structures.



[1] Toolbar

There are many buttons to control MAXIM. Detailed description and usage of each button are provided in succeeding sections.

[2] Progress bar

The computation progress is represented here during calculation.

[3] Timer

The computation time is indicated here after calculation.

[4] Mode tab

A simulation mode can be switched between a Normal and Sweep mode.

[5] Schedule

A simulation condition is displayed and modified here. The number of columns varies depending on the structures. Basic environmental variables (See Section 1.3) are represented as

- wavelength: Incident wavelength [nm]
- theta: Incident angle [°]
- phi: Angle phi [°]
- psi: Polarization angle [°]
- fx: Fourier truncation order in x direction
- fy: Fourier truncation order in y direction
- tx: Structure period [nm] in x direction
- ty: Structure period [nm] in y direction
- ni: Refractive index of input medium (ex) 2, 2+0.4j, SiO2 ...
- nf: Refractive index of output medium (ex) 2, 2+0.4j, SiO2 ...
- zmin: Additional simulation range [nm] in $-z$ direction
- zmax: Additional simulation range [nm] in $+z$ direction

[6] Field viewer

Electromagnetic field distribution is presented here. Users can check each data point by clicking the point.

[7] Status bar

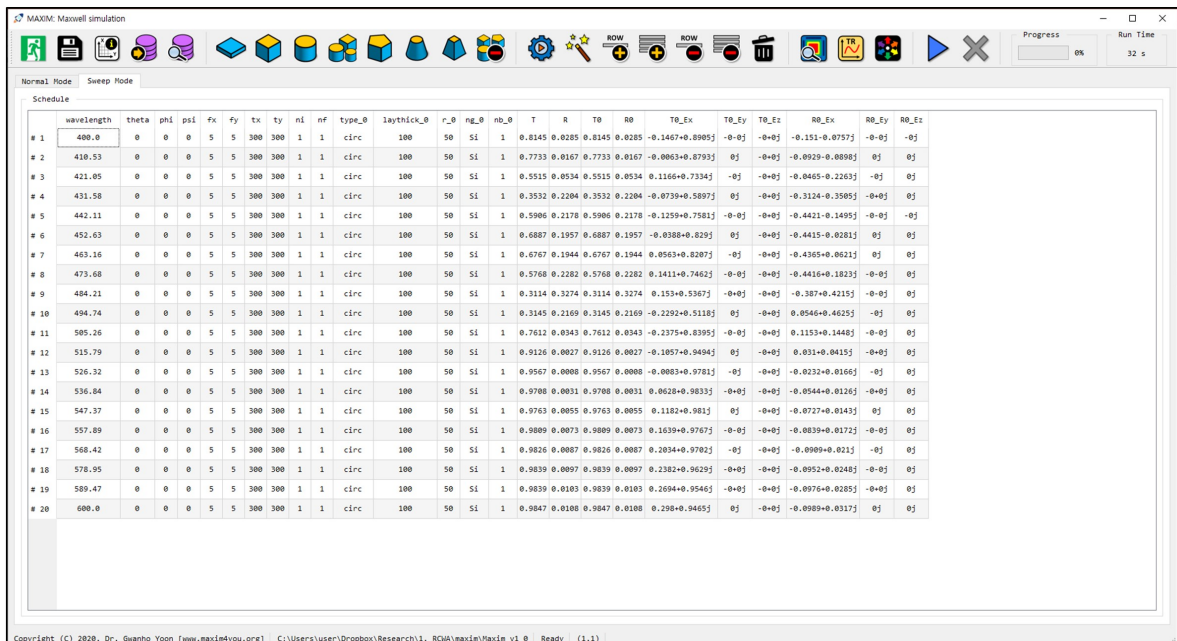
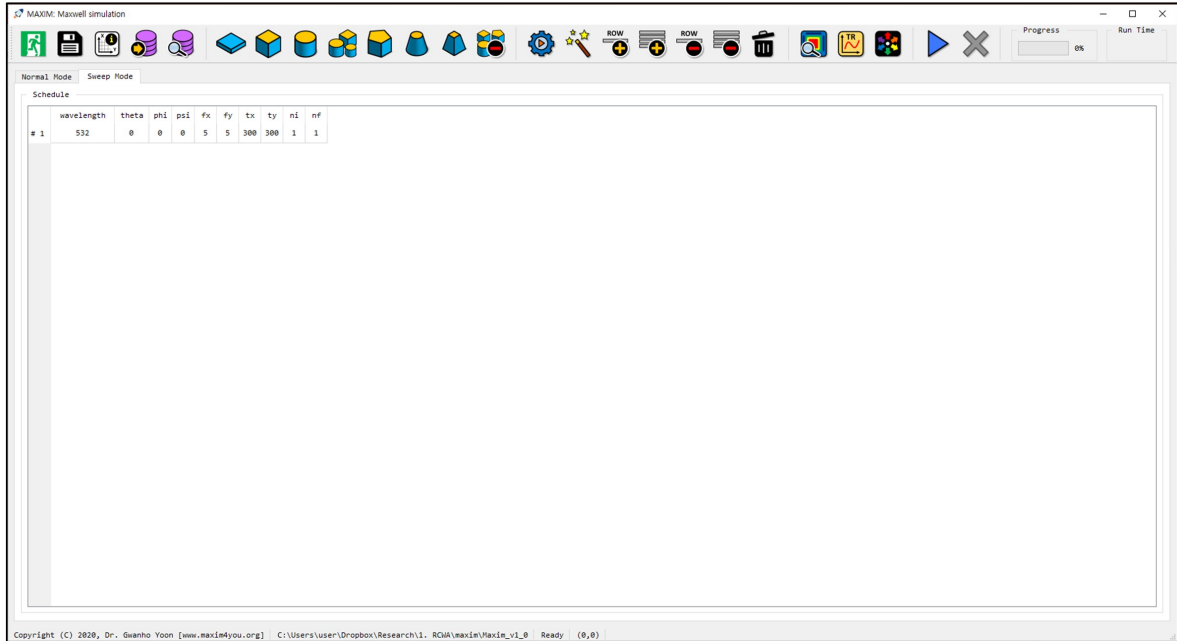
The current status is shown here.

[8] Result

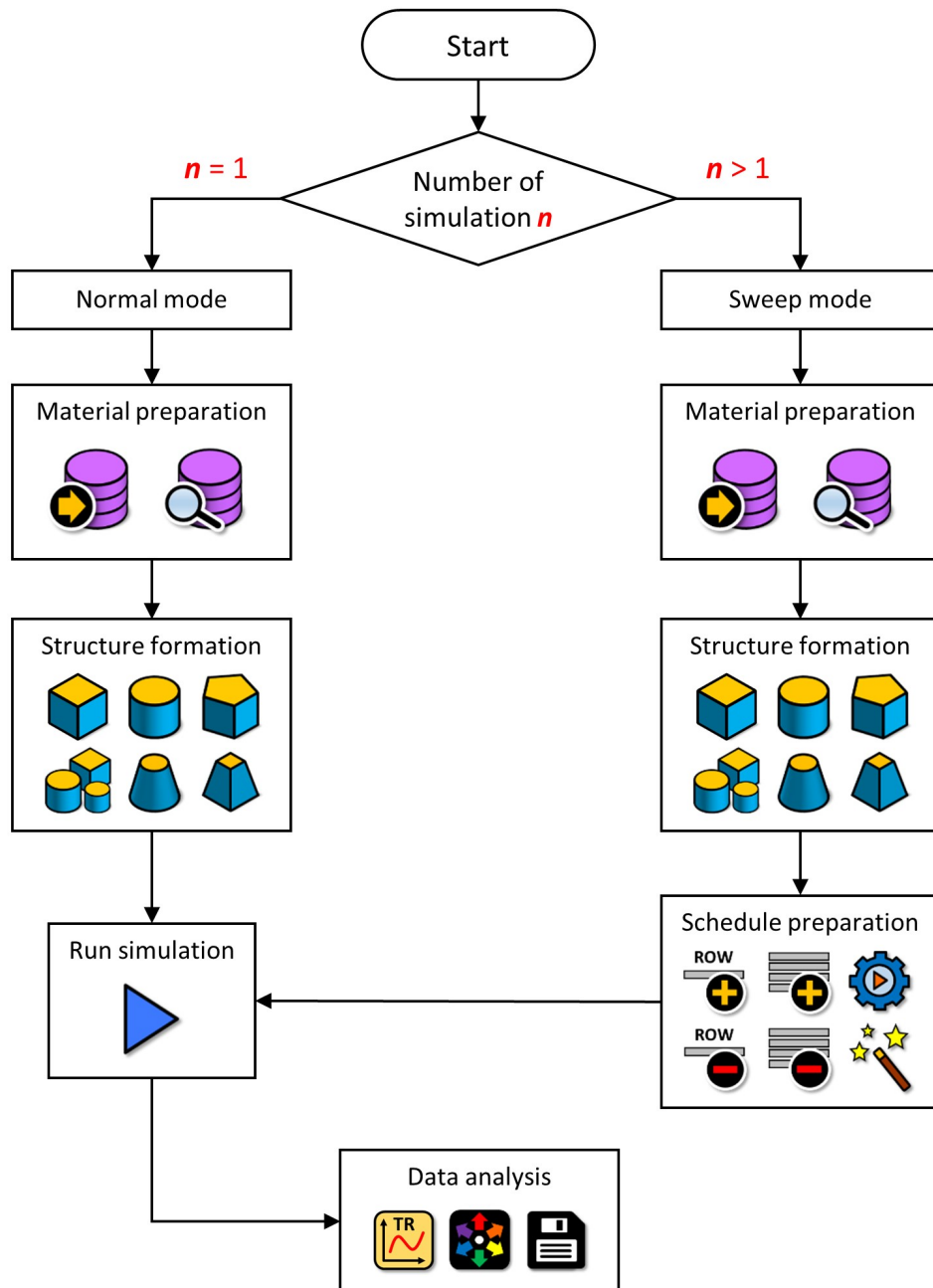
- T: Total transmittance
- R: Total reflectance
- T0: Transmittance of zeroth order diffraction
- R0: Reflectance of zeroth order diffraction
- T0_Ex: Zeroth order transmitted complex amplitude of electric field in x direction
- T0_Ey: Zeroth order transmitted complex amplitude of electric field in y direction
- T0_Ez: Zeroth order transmitted complex amplitude of electric field in z direction
- R0_Ex: Zeroth order reflected complex amplitude of electric field in x direction
- R0_Ey: Zeroth order reflected complex amplitude of electric field in y direction
- R0_Ez: Zeroth order reflected complex amplitude of electric field in z direction

<Sweep Mode>

In the Sweep mode, many simulation conditions with different parameters can be examined in series. Several toolbar buttons are activated to create a 2D schedule table which indeed can be managed in a very convenient way due to those functions.



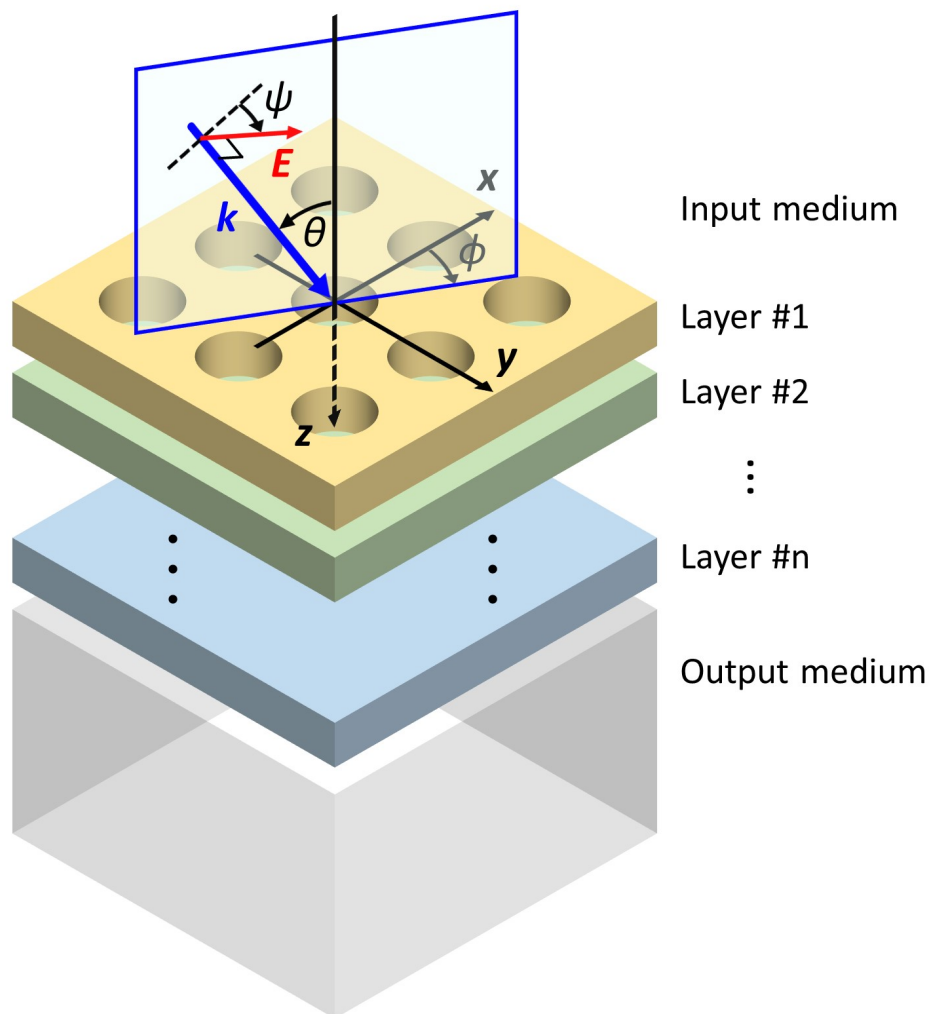
1.2 Basic Use



1.3 Coordinate Information

ICON	DESCRIPTION
	Display a coordinate schematic of MAXIM

This button displays a following coordinate schematic of MAXIM. Users should keep in mind the electric field vector is aligned into x direction when three angles (θ , ϕ , ψ) become zero.



1.4 Exit

ICON	DESCRIPTION
	Terminate MAXIM

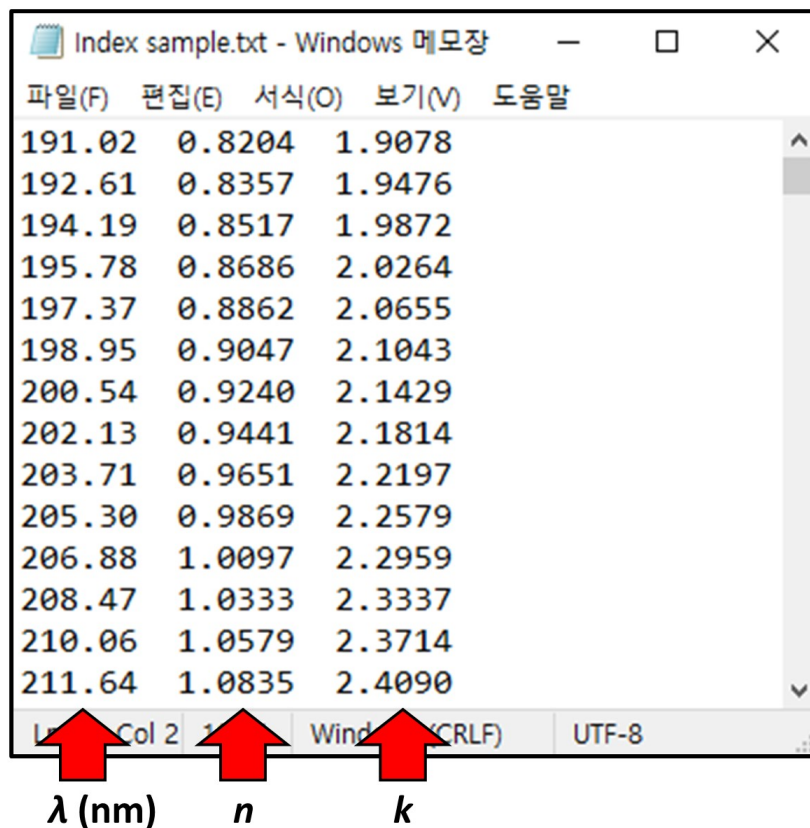
This button terminates MAXIM.

2 Materials

2.1 Material Import

ICON	DESCRIPTION
	Import new material data

This button enables to import new material data of refractive index. The allowed format of the data is a 3-column table where each column contains wavelength, refractive index and extinction coefficient, respectively. Each column should be separated by a TAB. The length unit of the wavelength is nanometer and the acceptable file format is a text file (.txt). In computation, missing data points of the refractive index for a given wavelength are obtained by linear interpolation of two adjacent data points automatically. The following figure shows an exemplary file of the material index data.

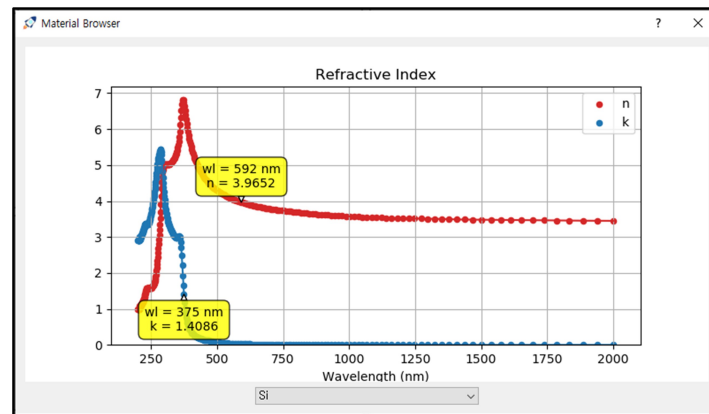


191.02	0.8204	1.9078
192.61	0.8357	1.9476
194.19	0.8517	1.9872
195.78	0.8686	2.0264
197.37	0.8862	2.0655
198.95	0.9047	2.1043
200.54	0.9240	2.1429
202.13	0.9441	2.1814
203.71	0.9651	2.2197
205.30	0.9869	2.2579
206.88	1.0097	2.2959
208.47	1.0333	2.3337
210.06	1.0579	2.3714
211.64	1.0835	2.4090

2.2 Material Browser

ICON	DESCRIPTION
	Browse the material data in MAXIM

This button display a refractive index plot of materials imported in MAXIM as follows. Users can check each data point by clicking the point. In computation, missing data points of the refractive index for a given wavelength are obtained by linear interpolation of two adjacent data points automatically.



MAXIM also provides refractive indices of many materials as follows.

NAME	SOURCE
Ag	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
Al	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
Al ₂ O ₃	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
AlN	J. Pastrnak and L. Roskovcova, Phys. Stat. Sol. 14, K5-K8 (1966)
Au	P. B. Johnson and R. W. Christy, Phys. Rev. B 6, 4370-4379 (1972)
GaN	A. S. Barker Jr. and M. Ilegems, Phys. Rev. B 7, 743-750 (1973)
Ge	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
HfO ₂	D. L. Wood, Kurt Nassau, T. Y. Kometani, and D. L. Nash, Appl. Opt. 29, 604-607 (1990)
LiF	H. H. Li, J. Phys. Chem. Ref. Data 5, 329-528 (1976)
MgF ₂	M. J. Dodge, Appl. Opt. 23, 1980-1985 (1984)
Si	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
Si ₃ N ₄	H. R. Philipp, J. Electrochim. Soc. 120, 295-300 (1973)
SiO ₂	Handbook of Optical Constants of Solids, Edward D. Palik, Academic Press, Boston, 1985
TiO ₂	Deposited and measured by Gwanho Yoon (2017)
aSi	D. T. Pierce and W. E. Spicer, Phys. Rev. B 5, 3017-3029 (1972)
aSiH	G. Yoon, J. Kim, J. Mun, D. Lee, K.T. Nam and J. Rho, Commun. Phys. 2, 129, (2019)

3 Structures

3.1 Film

ICON	DESCRIPTION
	Add film

This button creates the following additional cells in the table to add a film structure on the exiting structure.

<code>laythick_0</code>	<code>n_0</code>
-------------------------	------------------

`laythick`: Thickness [nm] of the film

`n`: Refractive index of the film (ex) 2, 2+0.4j, SiO2 ...

The subscriptions after headings represent the order of the structure.

3.2 Rectangle

ICON	DESCRIPTION
	Add rectangle

This button creates the following additional cells in the table to add a 2D rectangular structure on the exiting structure. It is worth noting that a 1D grating can be generated by setting either $fx = 0$ or $fy = 0$.

<code>type_0</code>	<code>laythick_0</code>	<code>wx_0</code>	<code>wy_0</code>	<code>dx_0</code>	<code>dy_0</code>	<code>ng_0</code>	<code>nb_0</code>
<code>rect</code>							

`laythick`: Thickness [nm] of the structure

`wx`: Length [nm] of the structure in x direction

`wy`: Length [nm] of the structure in y direction

`dx`: Center position [nm] of the structure in x direction

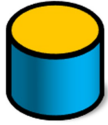
`dy`: Center position [nm] of the structure in y direction

`ng`: Refractive index of the structure (ex) 2, 2+0.4j, SiO2 ...

`nb`: Refractive index of the surrounding medium (ex) 2, 2+0.4j, SiO2 ...

The subscriptions after headings represent the order of the structure.

3.3 Circle

ICON	DESCRIPTION
	Add circle

This button creates the following additional cells in the table to add a cylindrical cavity on the exiting structure.

<code>type_0</code>	<code>laythick_0</code>	<code>r_0</code>	<code>ng_0</code>	<code>nb_0</code>
<code>circ</code>				

`laythick`: Thickness [nm] of the structure

`r`: Radius [nm] of the structure

`ng`: Refractive index of the structure (ex) 2, 2+0.4j, SiO2 ...

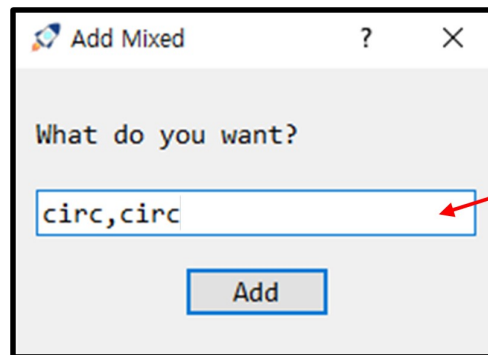
`nb`: Refractive index of the surrounding medium (ex) 2, 2+0.4j, SiO2 ...

The subscriptions after headings represent the order of the structure.

3.4 Mixed

ICON	DESCRIPTION
	Add multiple rectangles and circles

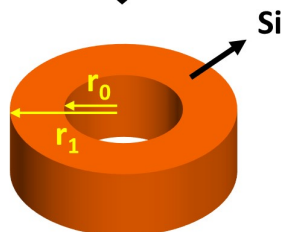
This button enables to add multiple 2D rectangular structures and cylindrical cavities in the unit cell simultaneously. Added structures can overlap each other and refractive indices of them can be set independently. Therefore, many kinds of structures including cross, ring and their combinations can be created. The following example could help to understand the usage of this function.



Enter **rect** or **circ**
separated by **comma(,)**



type_0	laythick_0	r0_0	dx0_0	dy0_0	ng0_0	r1_0	dx1_0	dy1_0	ng1_0	nb_0
mixed	100	50	0	0	1	100	0	0	Si	1



3.5 Polygon

ICON	DESCRIPTION
	Add polygons

This button enables to add multiple polygons in which cross-sections are defined by the coordinates of vertices in xy plane. Added structures can overlap each other and refractive indices of them can be set independently. The following example could help to understand the usage of this function.

?

×

Add Poly

How many polygons?

Vertices?

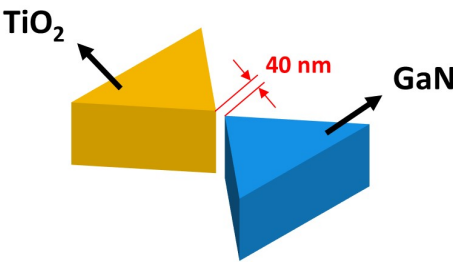
Add

Enter the number of polygons

Enter the number of vortices for each polygons separated by **comma(,)**



type_0	laythick_0	x0_0_0	y0_0_0	x1_0_0	y1_0_0	x2_0_0	y2_0_0	ng_0_0	x0_1_0	y0_1_0	x1_1_0	y1_1_0	x2_1_0	y2_1_0	ng_1_0	nb_0
poly	100	0	-20	100	-100	-100	-100	TiO2	0	20	100	100	-100	100	GaN	1



3.6 Cone

ICON	DESCRIPTION
	Add truncated cone

This button creates the following additional cells in the table to add a truncated cone on the exiting structure. It is worth noting that the cone is automatically divided into multiple layers of cylindrical cavities with varying radii by internal algorithms.

<code>type_0</code>	<code>laythick_0</code>	<code>r_in_0</code>	<code>r_out_0</code>	<code>lays_0</code>	<code>ng_0</code>	<code>nb_0</code>
<code>cone</code>						

`laythick`: Thickness [nm] of the structure

`r_in`: Radius [nm] of the structure where the light goes in

`r_out`: Radius [nm] of the structure where the light comes out

`lays`: The number of layers to approximate the structure

`ng`: Refractive index of the structure (ex) 2, 2+0.4j, SiO₂, TiO₂/SiO₂ ...

`nb`: Refractive index of the surrounding medium (ex) 2, 2+0.4j, SiO₂, TiO₂/SiO₂ ...

The subscriptions after headings represent the order of the structure.

3.7 Horn

ICON	DESCRIPTION
	Add rectangular horn

This button creates the following additional cells in the table to add a rectangular horn on the exiting structure. It is worth noting that the horn is automatically divided into multiple layers of rectangular structures with varying sizes by internal algorithms.

type_0	laythick_0	wx_in_0	wy_in_0	wx_out_0	wy_out_0	lays_0	ng_0	nb_0
horn								

laythick: Thickness [nm] of the structure

wx_in: Length [nm] of the structure in x direction where the light goes in

wy_in: Length [nm] of the structure in y direction where the light goes in

wx_out: Length [nm] of the structure in x direction where the light comes out

wy_out: Length [nm] of the structure in y direction where the light comes out

lays: The number of layers to approximate the structure

ng: Refractive index of the structure (ex) 2, 2+0.4j, SiO₂, TiO₂/SiO₂ ...

nb: Refractive index of the surrounding medium (ex) 2, 2+0.4j, SiO₂, TiO₂/SiO₂ ...

The subscriptions after headings represent the order of the structure.

3.8 Delete

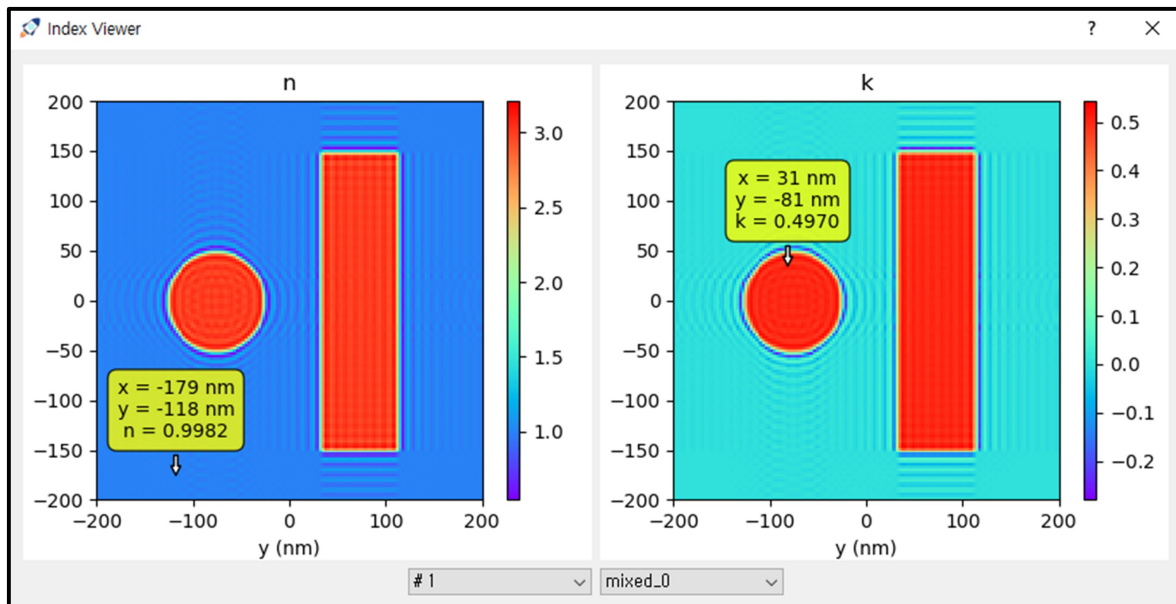
ICON	DESCRIPTION
	Delete a structure layer

This button removes one structure layer of the largest order.

3.9 Index Viewer

ICON	DESCRIPTION
	Display inplane index distribution of existing structures

This button shows the refractive index distribution of existing structures layer-by-layer as the following example ($f_x = f_y = 20$). Users can check each data point by clicking the point. The recovered index distribution approaches more closer to the original structures as the truncation order increases, but an acceptable compromise is required due to drastically increasing computation costs. It is recommended to use this function to check the validity of the truncation before the main RCWA computation.



4 Schedule

4.1 Basic Management

ICON	DESCRIPTION
ROW 	Add a row
	Add multiple rows
ROW 	Remove a row
	Remove multiple rows

These buttons activated in the Sweep mode enable to add or remove rows of the schedule table. A newly added row is filled with the values of the preceding row.

4.2 Auto Complete

ICON	DESCRIPTION
	Complete a schedule table automatically

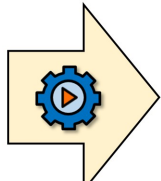
This button supports to complete and modify a column of the schedule table in a convenient manner. There are some rules in this function depending on the number of selected cells and following examples will help to catch the usage. It is worth noting that this function is valid for all columns.

Rules

- One cell: Duplication
- Two cells: Linear extrapolation
- Multiple cells: Linear interpolation

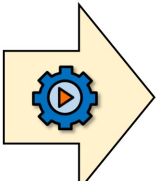
One cell

wavelength
500
0
0
0
0



wavelength
500
500
500
500
500

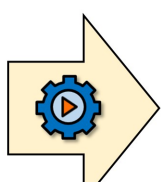
ni
SiO2
1
1
1
1



ni
SiO2
SiO2
SiO2
SiO2
SiO2

Two cells

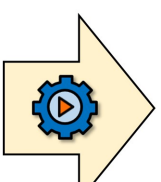
wavelength
500
600
0
0
0



wavelength
500.0
600.0
700.0
800.0
900.0

Multiple cells

wavelength
500
0
0
600
0



wavelength
500.0
533.33
566.67
600.0
0

4.3 Table Wizard

ICON	DESCRIPTION
	Create a programmed schedule table

This button provides a very convenient way to create a desired schedule table by using a built-in script. The arrangement level of each column increases as the column goes to the right. Following examples help to comprehend the operation of this fascinating function.

Scripting

[val1] , [val2] , [val3] , ... → Multiple values

[val1] : [val2] : [num] → Evenly spaced values over a specified interval

[header] → Duplication of the column specified by the header

[header] ; [val1] , [val2] , [val3] , ... → Parallel sweep with multiple values

[header] ; [val1] : [val2] : [num] → Parallel sweep with evenly spaced values

wavelength	theta	phi	psi	fx	fy	tx	ty	ni	nf
400,500,600	0	0:90:2	phi	phi;5,10	phi;5:10:2	wavelength	wavelength	SiO2,TiO2	1

wavelength	theta	phi	psi	fx	fy	tx	ty	ni	nf
400	0	0.0	0.0	5	5.0	400	400	SiO2	1
400	0	0.0	0.0	5	5.0	400	400	TiO2	1
400	0	90.0	90.0	10	10.0	400	400	SiO2	1
400	0	90.0	90.0	10	10.0	400	400	TiO2	1
500	0	0.0	0.0	5	5.0	500	500	SiO2	1
500	0	0.0	0.0	5	5.0	500	500	TiO2	1
500	0	90.0	90.0	10	10.0	500	500	SiO2	1
500	0	90.0	90.0	10	10.0	500	500	TiO2	1
600	0	0.0	0.0	5	5.0	600	600	SiO2	1
600	0	0.0	0.0	5	5.0	600	600	TiO2	1
600	0	90.0	90.0	10	10.0	600	600	SiO2	1
600	0	90.0	90.0	10	10.0	600	600	TiO2	1

4.4 Reset

ICON	DESCRIPTION
	Initialize a schedule table

This button removes all the structures and rows.

5 Execution

5.1 Run

ICON	DESCRIPTION
	Run scheduled simulation

This button runs simulation.

5.2 Cancellation

ICON	DESCRIPTION
	Force quit simulation

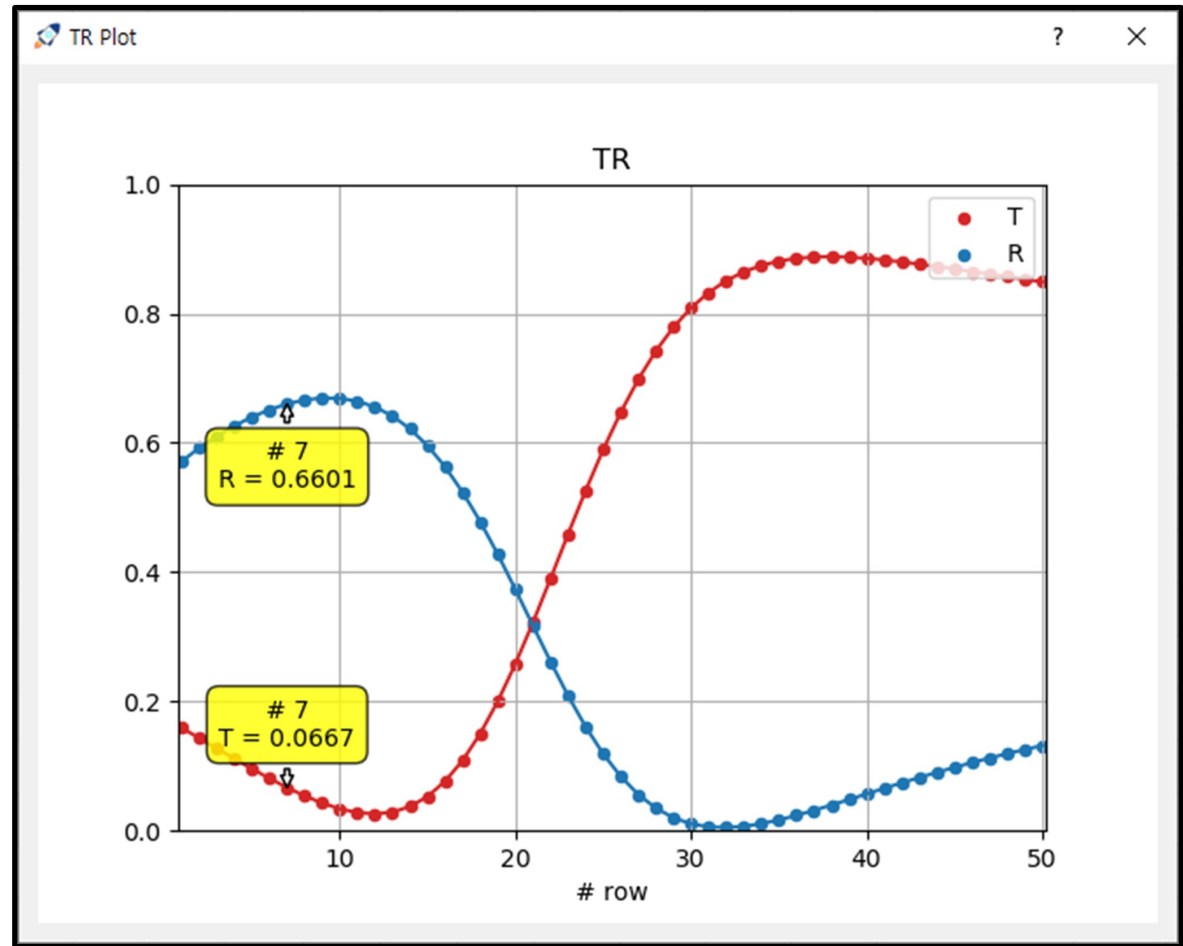
This button can force quit the main simulation and calculation of Index Viewer.

6 Result

6.1 Transmittance & Reflectance

ICON	DESCRIPTION
	Display calculated transmittance and reflectance

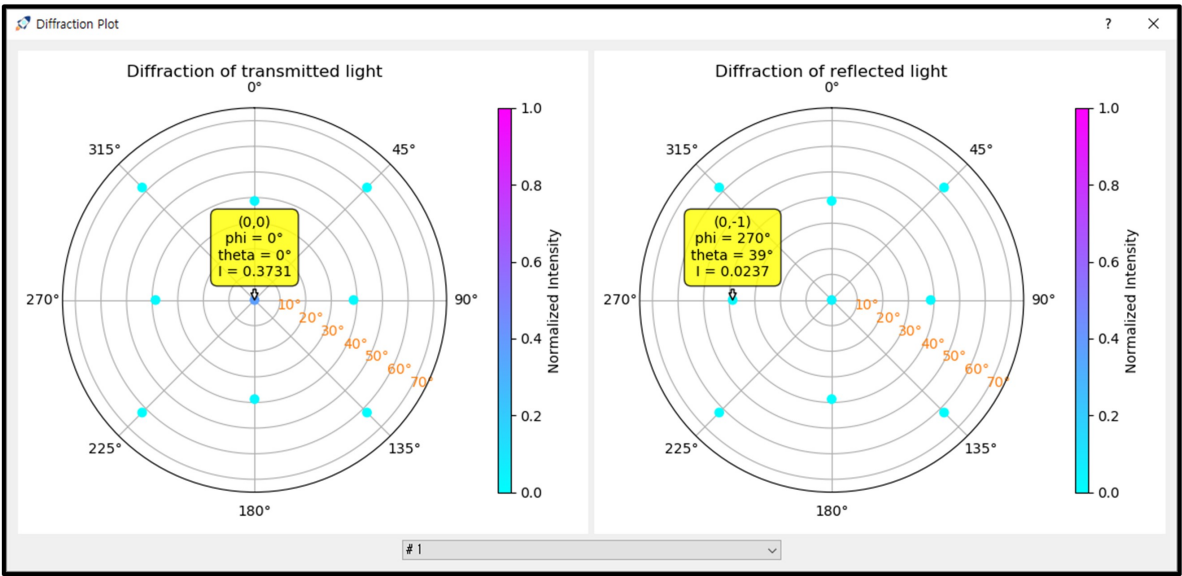
This button displays a plot of calculated total transmittance and reflectance as follows. Users can check each data point by clicking the point. It is recommended to use this function in the Sweep mode due to its sufficiently large number of data points.



6.2 Diffraction Analysis

ICON	DESCRIPTION
	Display calculated diffraction angles and corresponding powers

This button presents diffraction angles and powers in both input and output media as follows. Users can check each data point by clicking the point. The first line of the annotation boxes indicates diffraction orders in x and y direction, respectively while θ and ϕ correspond to those represented in the coordinate information. In the Sweep mode, it is also possible to display the result row-by-row using a combo box below the plots.



6.3 Data Export

ICON	DESCRIPTION
	Save calculation results

This button exports the calculation results into a designated folder. The exported data contain a result table, discrete wavevector components and complex coefficients of transmission and reflection. In the Normal mode, electromagnetic field data are also exported together. The file extension of the result table is CSV while the rest of data have MAT file extension compatible to other computation software such as MATLAB.

7 Information

7.1 Packages

MAXIM utilizes several open-source packages to compute RCWA formula, display computation results and build GUIs. All the packages used in MAXIM is presented in the following list.

NAME	DESCRIPTION
Python 3.8	A basic programming language
PySide2	A GUI package including table, button and window
Numpy	Mathematical functions of linear algebra
Scipy	Mathematical functions of linear algebra
Pandas	Functions to create and manipulate tables
Matplotlib	Graphical display functions such as plotting
Mplcursors	Annotation functions based on a cursor
Mpldatacousor	Annotation functions based on a cursor
Pyinstaller	A package converting Python applications to stand-alone executables

7.2 License

GNU LESSER GENERAL PUBLIC LICENSE

The MAXIM is Copyright (C) 2020 Gwanho Yoon

Contact: <https://ylab.seoultech.ac.kr>

You may use, distribute and copy the MAXIM under the terms of GNU Lesser General Public License version 3, which is displayed below.

GNU LESSER GENERAL PUBLIC LICENSE Version 3, 29 June 2007

Copyright (C) 2007 Free Software Foundation, Inc. (<http://fsf.org/>) Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

This version of the GNU Lesser General Public License incorporates the terms and conditions of version 3 of the GNU General Public License, supplemented by the additional permissions listed below.

0. Additional Definitions.

As used herein, "this License" refers to version 3 of the GNU Lesser General Public License, and the "GNU GPL" refers to version 3 of the GNU General Public License.

"The Library" refers to a covered work governed by this License, other than an Application or a Combined Work as defined below.

An "Application" is any work that makes use of an interface provided by the Library, but which is not otherwise based on the Library. Defining a subclass of a class defined by the Library is deemed a mode of using an interface provided by the Library.

A "Combined Work" is a work produced by combining or linking an Application with the Library. The particular version of the Library with which the Combined Work was made is also called the "Linked Version".

The "Minimal Corresponding Source" for a Combined Work means the Corresponding Source for the Combined Work, excluding any source code for portions of the Combined Work that, considered in isolation, are based on the Application, and not on the Linked Version.

The "Corresponding Application Code" for a Combined Work means the object code and/or source code for the Application, including any data and utility programs needed for reproducing the Combined Work from the Application, but excluding the System Libraries of the Combined Work.

1. Exception to Section 3 of the GNU GPL.

You may convey a covered work under sections 3 and 4 of this License without being bound by section 3 of the GNU GPL.

2. Conveying Modified Versions.

If you modify a copy of the Library, and, in your modifications, a facility refers to a function or data to be supplied by an Application that uses the facility (other than as an argument passed when the facility is invoked), then you may convey a copy of the modified version:

- a) under this License, provided that you make a good faith effort to ensure that, in the event an Application does not supply the function or data, the facility still operates, and performs whatever part of its purpose remains meaningful, or
- b) under the GNU GPL, with none of the additional permissions of this License applicable to that copy.

3. Object Code Incorporating Material from Library Header Files.

The object code form of an Application may incorporate material from a header file that is part of the Library. You may convey such object code under terms of your choice, provided that, if the incorporated material is not limited to numerical parameters, data structure layouts and accessors, or small macros, inline functions and templates (ten or fewer lines in length), you do both of the following:

- a) Give prominent notice with each copy of the object code that the Library is used in it and that the Library and its use are covered by this License.
- b) Accompany the object code with a copy of the GNU GPL and this license document.

4. Combined Works.

You may convey a Combined Work under terms of your choice that, taken together, effectively do not restrict modification of the portions of the Library contained in the Combined Work and reverse engineering for debugging such modifications, if you also do each of the following:

- a) Give prominent notice with each copy of the Combined Work that the Library is used in it and that the Library and its use are covered by this License.
- b) Accompany the Combined Work with a copy of the GNU GPL and this license document.
- c) For a Combined Work that displays copyright notices during execution, include the copyright notice for the Library among these notices, as well as a reference directing the user to the copies of the GNU GPL and this license document.
- d) Do one of the following:
 - 0) Convey the Minimal Corresponding Source under the terms of this License, and the Corresponding Application Code in a form suitable for, and under terms that permit, the user to recombine or relink the Application with a modified version of the Linked Version to produce a modified Combined Work, in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.
 - 1) Use a suitable shared library mechanism for linking with the Library. A suitable mechanism is one that (a) uses at run time a copy of the Library already present on the user's computer system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.
- e) Provide Installation Information, but only if you would otherwise be required to provide such information under section 6 of the GNU GPL, and only to the extent that such information is necessary to install and execute a modified version of the Combined Work produced by recombining or relinking the Application with a modified version of the Linked Version. (If you use option 4d0, the Installation Information must accompany the Minimal Corresponding Source and Corresponding Application Code. If you use option 4d1, you must provide the Installation Information in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.)

5. Combined Libraries.

You may place library facilities that are a work based on the Library side by side in a single li-

brary together with other library facilities that are not Applications and are not covered by this License, and convey such a combined library under terms of your choice, if you do both of the following:

- a) Accompany the combined library with a copy of the same work based on the Library, uncombined with any other library facilities, conveyed under the terms of this License.
- b) Give prominent notice with the combined library that part of it is a work based on the Library, and explaining where to find the accompanying uncombined form of the same work.

6. Revised Versions of the GNU Lesser General Public License.

The Free Software Foundation may publish revised and/or new versions of the GNU Lesser General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Library as you received it specifies that a certain numbered version of the GNU Lesser General Public License "or any later version" applies to it, you have the option of following the terms and conditions either of that published version or of any later version published by the Free Software Foundation. If the Library as you received it does not specify a version number of the GNU Lesser General Public License, you may choose any version of the GNU Lesser General Public License ever published by the Free Software Foundation.

If the Library as you received it specifies that a proxy can decide whether future versions of the GNU Lesser General Public License shall apply, that proxy's public statement of acceptance of any version is permanent authorization for you to choose that version for the Library.